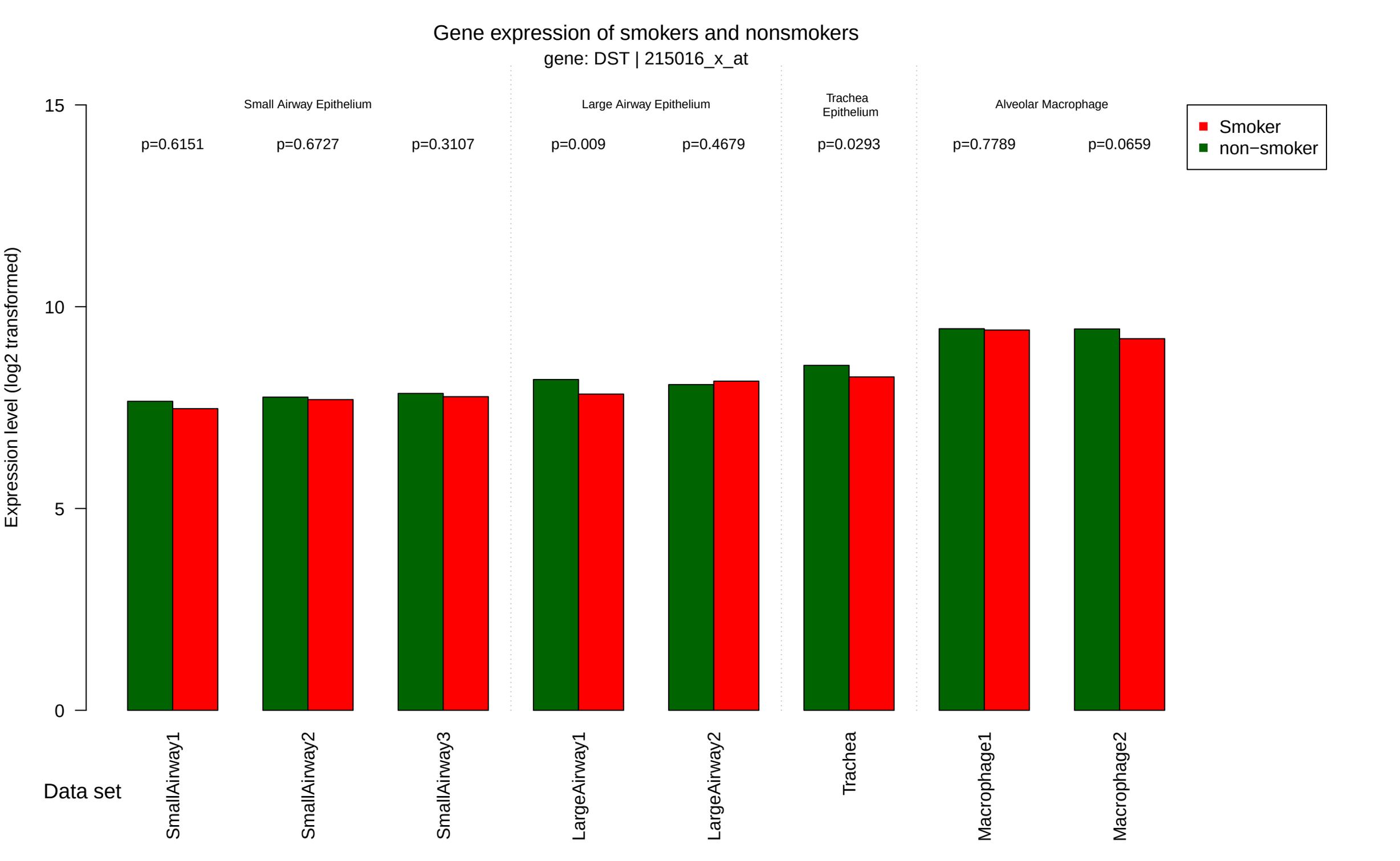
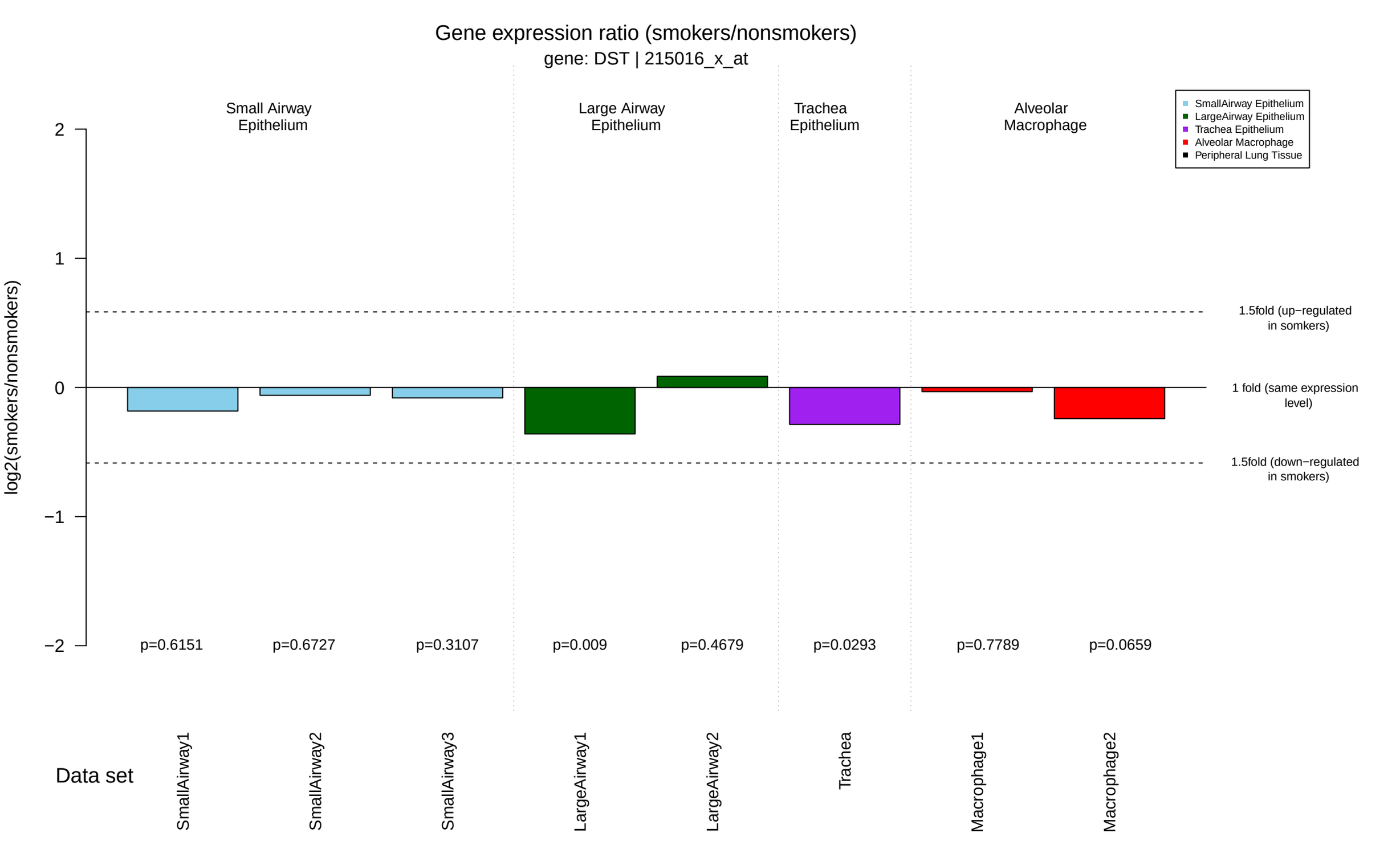
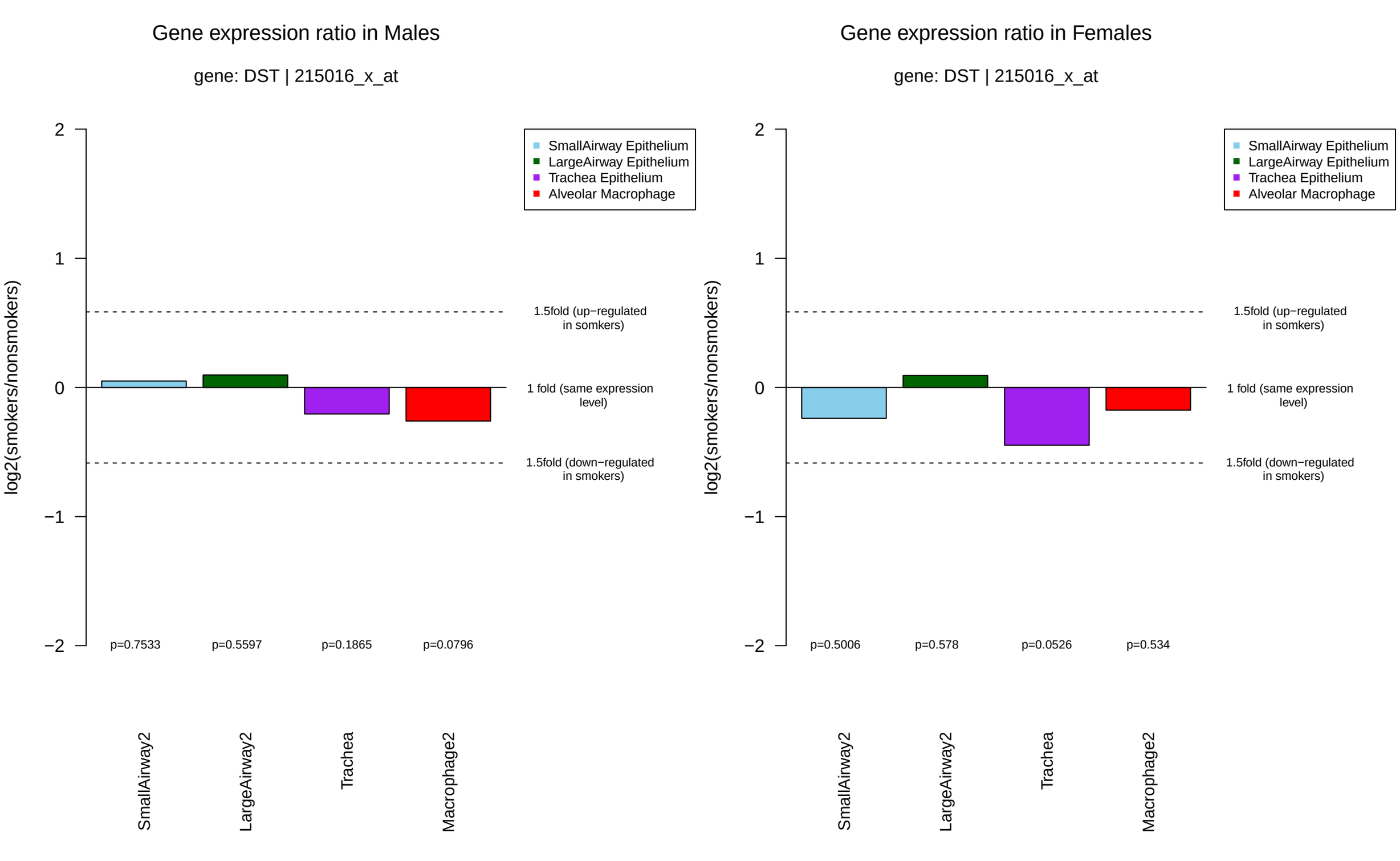
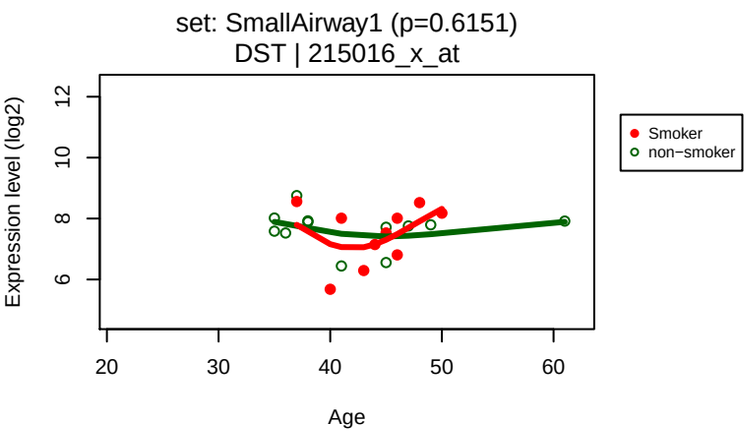
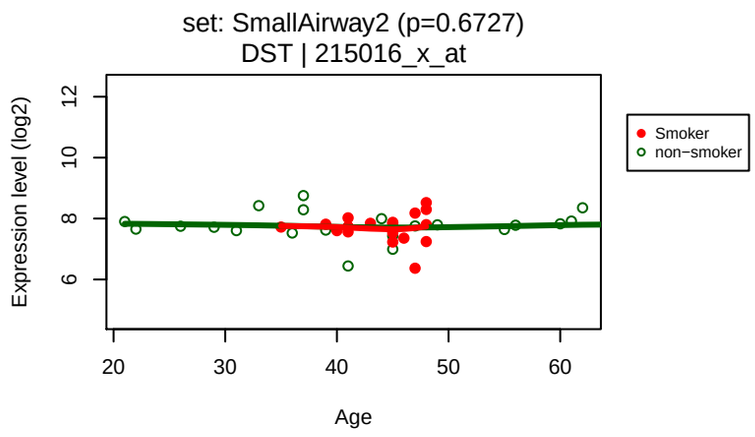
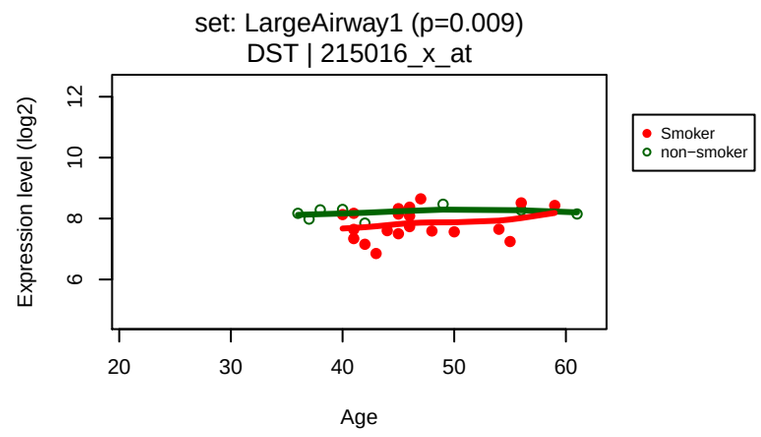

Gene expression of smokers and nonsmokers
gene: DST | 215016_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6151
p=0.6727
p=0.3107
p=0.009
p=0.4679
p=0.0293
p=0.7789
p=0.0659
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

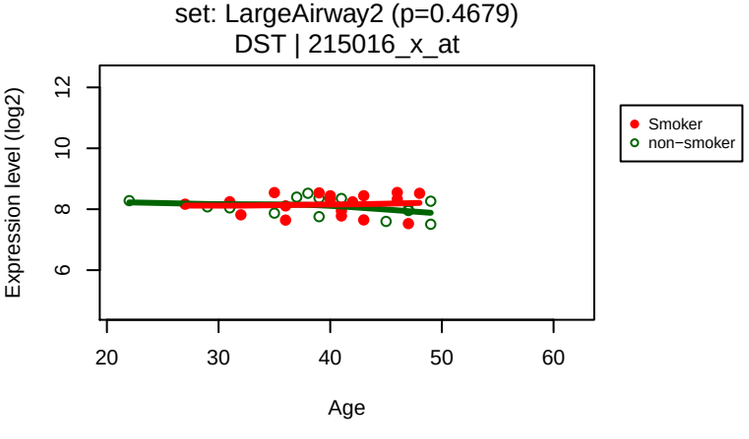

Gene expression ratio (smokers/nonsmokers)
gene: DST | 215016_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6151
p=0.6727
p=0.3107
p=0.009
p=0.4679
p=0.0293
p=0.7789
p=0.0659
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

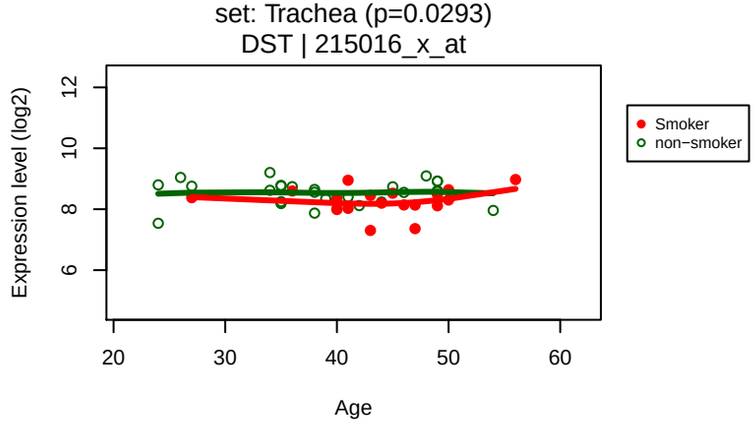

Gene expression ratio in Males
gene: DST | 215016_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7533
p=0.5597
p=0.1865
p=0.0796
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DST | 215016_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5006
p=0.578
p=0.0526
p=0.534
SmallAirway2
LargeAirway2
Trachea
Macrophage2

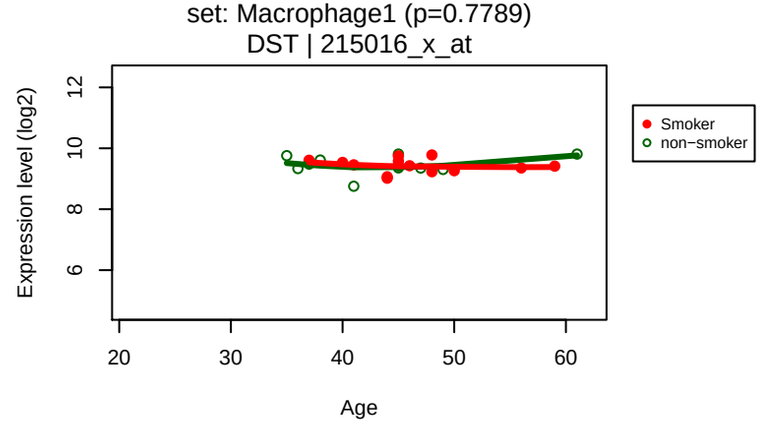

set: SmallAirway1 (p=0.6151)
DST | 215016_x_at
Expression level (log2)
Age
Smoker
non−smoker

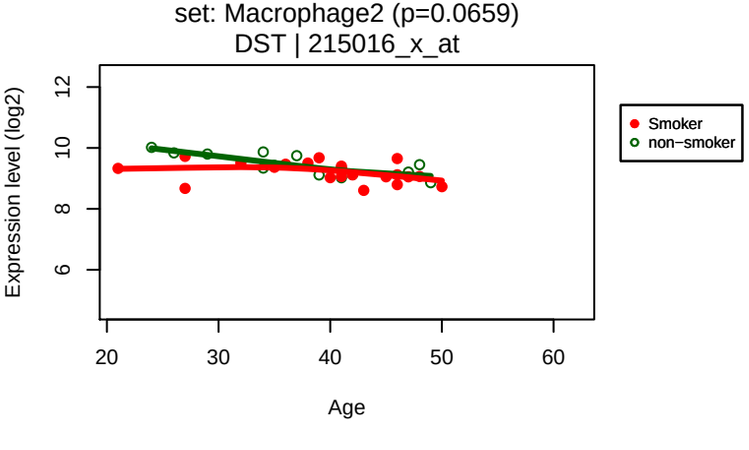

set: SmallAirway2 (p=0.6727)
DST | 215016_x_at
Expression level (log2)
Age
Smoker
non−smoker

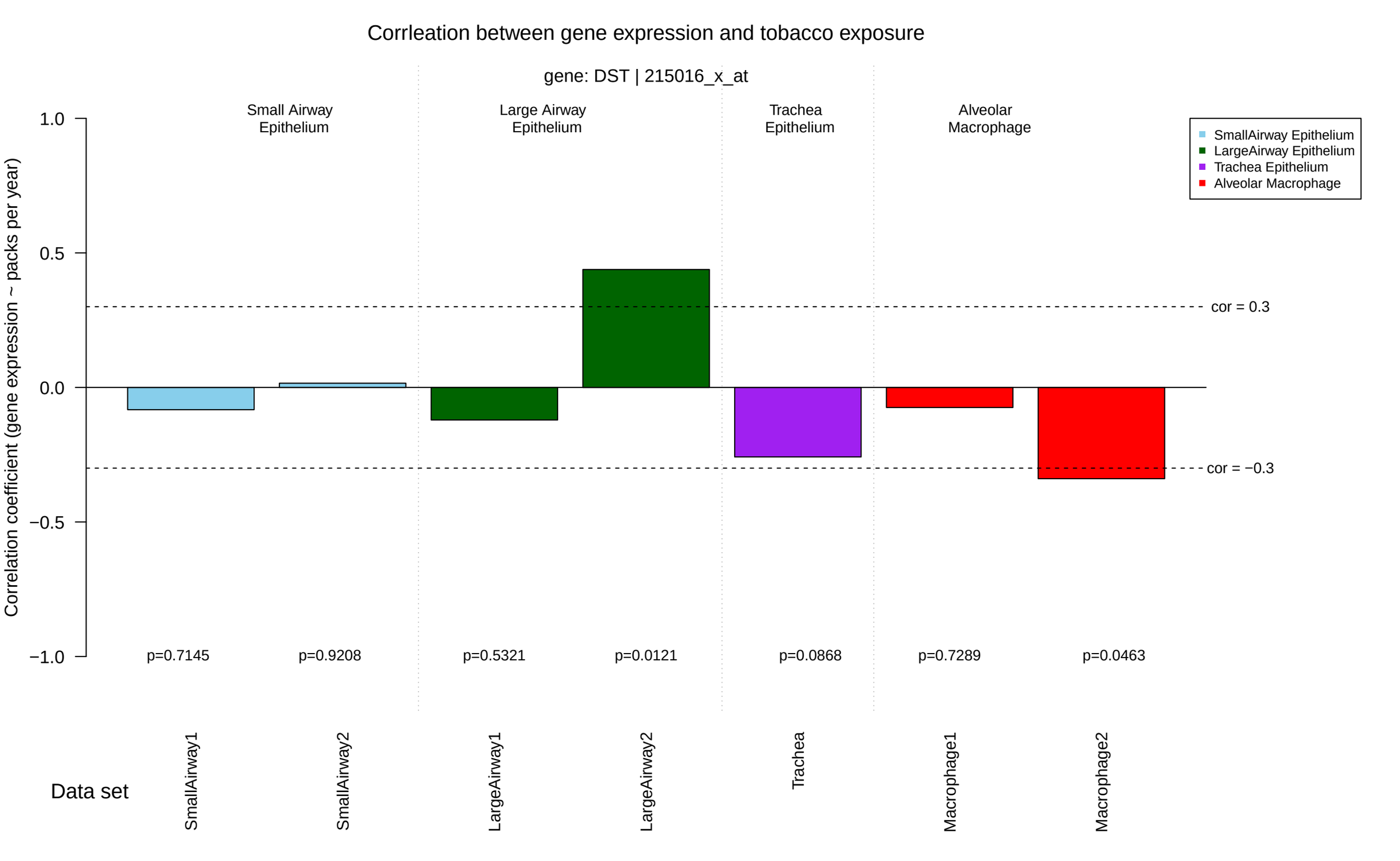

set: LargeAirway1 (p=0.009)
DST | 215016_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4679)
DST | 215016_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0293)
DST | 215016_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7789)
DST | 215016_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0659)
DST | 215016_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DST | 215016_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7145
p=0.9208
p=0.5321
p=0.0121
p=0.0868
p=0.7289
p=0.0463
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2